

Do Fresh Leadership 8-Ks Reward Put Sellers?

Gator Quant Hacks 2026 | Track 03: Systematic Trading & Massive 8-K Challenge | Strategy Quant Note | October 4, 2026

DECISION: DO NOT DEPLOY PROPOSED SIGNAL — We pre-registered a systematic test and found that fresh leadership-change filings generated no robust advantage for cash-secured put sellers over ordinary days. The hypothesis is decisively rejected: put sellers underperformed ordinary days across all 18 parameter combinations in-sample and out-of-sample. This unsupported payoff falsifies the trade; it does not prove buying puts would profit.

IN-SAMPLE EDGE (F-O)	REPORTED OOS (F-O)	PRIMARY CONTRAST (F-S)	VRP MAP SCREEN
-1.19% (n=55)	-0.70% (n=25)	-1.03% (n=55)	0 / 18 Pass (q=0.10)
Headline n=55 Excludes 0 (1/3)	Headline n=25 Spans 0 (0/3)	Headline n=55 Spans 0 (0/3)	FDR q=0.10 All CIs cross 0

01 SUMMARY

We pre-registered a systematic test of whether fresh leadership 8-Ks create a tradeable edge for selling downside insurance (cash-secured puts) due to temporary market overreaction. The centerpiece strategy enters a post-filing cash-secured put (CSP) struck 5% out-of-the-money (OTM) with 90-180 calendar days to expiration (targeting 120 days) at session close. The pre-registered hypothesis is decisively rejected: put sellers underperformed ordinary days in-sample (-1.19%, n=55) and in the project-reported 2026 out-of-sample window (-0.70%, n=25). The pre-outcome primary contrast (fresh minus stale filings) returned -1.03% in-sample and was not statistically significant. Across all 18 parameter neighbor combinations in both windows, the edge was consistently negative. Options markets do not overprice jump risk for fresh governance events; underlying stocks suffer persistent downward drift, and put sellers face uncompensated left-tail drawdowns.

02 ECONOMIC HYPOTHESIS

When an unplanned C-suite disruption occurs (CEO/CFO departure or appointment), investors face acute uncertainty. Microstructure theory suggests equity holders, surprised by an unexpected transition, overpay for downside insurance. If implied volatility exceeds realized post-filing volatility, put sellers should capture an attractive volatility risk premium. Conceptually, a short put decomposes into short Vega (harvesting implied volatility crush post-disclosure) and long Delta (directional stock exposure); the thesis required Vega capture to dominate Delta drag. Crucially, we hypothesize this overpricing exists only when the 8-K represents the first public disclosure ("fresh"). If an 8-K merely formalizes an executive transition announced days earlier via press release ("stale"), the chain has already re-priced. Pre-registered failure conditions in runs/PREREG.md specify that if fresh filings fail to beat ordinary days or perform indistinguishably from stale filings, the thesis is falsified. The counterparty is envisioned as an institutional hedger overpaying for puts; however, counterparty identities are unobserved.

03 DATA & UNIVERSE

The study uses the official challenge universe of the top 100 US equities by market capitalization (approximately the S&P 100) provided by Massive in Cell 11 of the starter notebook. Event disclosures are drawn from Massive Filings & Disclosures covering five Item 5.02 governance tags: ceo_appointment, ceo_departure, cfo_appointment, cfo_departure, and executive_officer_appointment. Event timing strictly respects market hours with zero backtest lookahead: filings accepted after 16:00 ET via EDGAR acceptance timestamps move to session t+1, and entry occurs strictly at that session's close. Freshness is determined by trading sessions elapsed between the SEC EDGAR CONFORMED PERIOD OF REPORT (date of earliest event reported) and the adjusted filing session: fresh if lag <= 1 trading session; stale if lag >= 2. Options pricing uses Massive daily option bars, contracts, and underlying equity references. Spot prices are synthetically inferred from ATM put-call parity. Only standard contracts are traded (post-split contracts dropped), and marks older than 3 sessions are discarded. In-sample event dates cover 2024-2025; the already-reported historical OOS window covers January-August 2026.

04 METHODOLOGY

Trade Construction: The primary instrument is a cash-secured put struck 5% below parity spot (OTM=0.05), selected from the 3-6 month expiry bucket (90-180 calendar days; 120-day target), entered at the close of the filing session. Positions are 100% cash-collateralized (strike x 100). Returns are measured at 9 fixed horizons: 1, 2, 3, 5, 10, 21, 42, 63 sessions, and contract expiry. The pre-registered headline equally averages differences at 21 sessions, 42 sessions, and expiry; it is neither an absolute trade return nor an annualized return.

Control Baseline (Placebo): The comparator uses same-ticker ordinary trading sessions randomly sampled at least 30 calendar days away from any SEC filing event (120 placebo draws per window). Event excess returns are defined strictly as event return minus ordinary-day control return. All reported differences represent percentage points of spot notional.

Statistical Inference: The original difference calculation independently resamples each group 2,000 times (seed 1) for percentile 95% bootstrap confidence intervals. The resampling is not issuer-clustered; observations can overlap. The hypothesis precedes recorded outcomes, but protocol date ranges were corrected after OOS; exact-window preregistration and independent tag custody are not established.

The 13-Experiment Research Funnel & Staged Gates: Research progressed through 13 structured tests, moving from measurement gates to economic evaluation:

Experiment / Stage	Design / Strategy	Sample Size / Denom.	Result / Gate	Key Finding / Methodological Takeaway
Phase 1-3 Date Gate	Excerpt date parsing	218 events (102 undated)	Gate Failed	46.8% undated; agreement 0.41. Replaced with EDGAR period-of-report.
P / A1 / A3 Leadership	Low-JEV leadership CSP / CC	31 to 150 priced events	Null / Fragile	Edges -0.53% to -1.78%. High-JEV arm empty; appointments mixed.
A2 / A4 / B1-B4 Arms	High-JEV exit, breach, collar	0 to 16 events in arm	Sparse Feasibility	Demoted A->B->C due to sparse high-JEV sample sizes.
B5 Good-News Call	Positive-tone tags / long call	103 priced events	Null (-0.15%)	Construct mismatch: tag labels do not imply positive price direction.
B6 Earnings Play	Earnings release / CSP and CC	117-118 priced IS events	Failed OOS	CSP +0.39% IS -> -0.46% OOS; CC +0.83% IS -> -1.59% OOS. Failed.
F1 Fresh Leadership	EDGAR lag <= 1 / CSP 120d	55 IS / 25 OOS priced	Claim Rejected	Fresh-vs-ord -1.19% IS / -0.70% OOS; fresh-minus-stale -1.03% IS.
VRP 18-Category Map	Implied vs realized move map	18 categories, 400 controls	Null Map	0/18 pass BH q=.10; all 95% CIs cross 0. H2 pooled OOS not confirmed.
Exp 6 Earnings Benchmark	Earnings resolution CSP +5	54 events / 17 CIK clusters	Candidate Failed	Paired net edge +0.0881% below +0.5% hurdle; failed 9 gates.
Exp 9B Uncertainty CSP	Governance resolution CSP +21	7 events / 7 issuers / 20 ctrl	Coverage Failure	Net edge -0.1728%; failed 20/10 event/issuer inference floor.
Exp 10 Covered Call	Governance resolution CC +21	6 events / 6 issuers / 17 ctrl	Coverage Failure	Net edge -0.8991%; failed 20/10 floor; all 36 sensitivity cells negative.
Exp 11-13 Forward Program	Adverse current x intact fwd	435 filings -> 12 signals	Infeasible (Rarity)	Expanded across 22+10 tags; 12 signals from 6 issuers. Extreme rarity.

Table 0: Structured progression of all 13 research stages from measurement gates to economic options testing.

05 RESULTS

5.1 Three-Horizon Headline Contrasts

The pre-registered primary contrast is fresh minus stale filings; fresh versus ordinary days is reported separately as a headline comparison. In-sample, the fresh cash-secured put underperformed ordinary days by -1.19% of spot notional (n=55), with the confidence interval excluding zero at 1 of 3 horizons. The pre-outcome primary contrast (fresh minus stale) returned -1.03% (n=55), with its 95% interval including zero across all three horizons (gate failed). In reported OOS, fresh versus ordinary returned -0.70% (n=25), repeating the negative sign but with intervals spanning zero at all three horizons. Stale versus ordinary returned +2.13% OOS, flipping sign from in-sample (-0.16%).

Window	Contrast	Gross diff.	Max N(a)	CI excludes 0
IS	Fresh - ordinary	-1.19%	55	1/3
IS	Stale - ordinary	-0.16%	95	0/3
IS	Fresh - stale (primary)	-1.03%	55	0/3
Reported OOS	Fresh - ordinary	-0.70%	25	0/3
Reported OOS	Stale - ordinary	+2.13%	40	2/3
Reported OOS	Fresh - stale (primary)	-2.82%	25	2/3

Table 1: Three-horizon gross differences and denominators. Max N(a) is headline maximum valid group count, not issuer N or matched set. Discovery counts: 62/108 IS, 27/42 OOS. Control-valid N, issuer N, matched N, and numeric F1 CI endpoints are unavailable. Source: ledger rows 87-89, 95-97.

5.2 All Fixed-Horizon Gross Differences

All 54 rounded gross differences below were recovered directly from saved original F1 notebook aggregate outputs at commit 5871597e3ecab5e0dbbc55d80314e1939d182224, cells 44 and 45. Underperformance deepens monotonically as the trade matures: in-sample fresh versus ordinary edge widens from -0.17% at session 1 to -0.65% at session 5, -1.03% at session 42, and -1.99% at expiry. In reported OOS, fresh underperformance reaches -1.67% at session 42 and -1.55% at session 63.

Sessions	F-O IS	S-O IS	F-S IS	F-O OOS	S-O OOS	F-S OOS
1	-0.17%	-0.30%*	+0.13%	+0.00%	+0.12%	-0.12%
2	-0.31%	-0.02%	-0.29%	-0.08%	+0.15%	-0.23%
3	-0.31%	-0.07%	-0.24%	+0.05%	+0.03%	+0.03%
5	-0.65%*	-0.24%	-0.41%	-0.01%	+0.22%	-0.23%
10	-0.88%	-0.23%	-0.65%	-0.25%	+0.28%	-0.54%
21	-0.56%	-0.28%	-0.28%	-0.58%	+0.64%	-1.21%
42	-1.03%	-0.04%	-0.99%	-1.67%	+1.88%*	-3.55%*
63	-1.63%	+0.02%	-1.65%	-1.55%	+2.14%*	-3.69%*
Expiry	-1.99%*	-0.16%	-1.83%	+0.16%	+3.86%*	-3.70%*

Table 2: All 54 fixed-horizon gross differences (% of spot notional). F-O: fresh-ordinary; S-O: stale-ordinary; F-S: fresh-stale (primary). * indicates original 95% bootstrap CI excludes zero; numeric endpoints and net differences remain unavailable.

5.3 Movement Diagnostics & Remote Benchmark Results

Directional move diagnostics explain the put seller's losses: underlying equities fell significantly more after fresh events than on ordinary days (own-stock move edge -1.32% IS, -1.36% OOS). Realized-to-implied move ratios show that fresh events moved 1.08x implied moves IS (vs 0.75x ordinary), but only 0.82x OOS (vs 0.65x ordinary). Remote priced benchmarks confirmed this lack of harvestable premium: Experiment 6 (earnings CSP) delivered a net edge of only +0.0881% at session +5, failing 9 gates; Experiment 9B (governance CSP) yielded a negative net edge of -0.1728%; and Experiment 10 (covered call) yielded -0.8991%.

5.4 Parameter Sensitivity & Multi-Category VRP Map

Robustness across parameter neighbors: All 18 reported cells (expiry bucket x moneyness x post/pre entry) in both in-sample and out-of-sample windows are negative. At 3-6m post-entry, edges across 3%, 5%, and 10% OTM are -1.32%, -1.19%*, -1.03% IS and -0.25%, -0.70%, -0.27%* OOS. Dropping the top 3 best IS events leaves a -0.45% edge. The negative result is a broad parameter plateau, not an artifact of 5% OTM strike selection.

Multi-Category Variance Risk Premium Map: Across 18 eligible 8-K categories, event-versus-ordinary mean $\log(|\text{realized}| / \text{implied move})$ was evaluated under Benjamini-Hochberg FDR control at $q=0.10$. Zero of 18 categories passed; every 95% confidence interval crosses zero. The H2 generalization test (fresh vs stale outside leadership tags) yielded diff +0.003 IS ($p=.999$) and -0.149 OOS ($p=.114$), confirming no generalized harvestable variance mispricing.

06 RISK MANAGEMENT

6.1 Simulated Portfolio Performance & Downside Metrics

We evaluated a 21-session simulated portfolio allocating 10% capital per event, charging a modeled 5% premium haircut per side (median 13 bps IS, 18 bps OOS; doubled for 2x cost stress). Sizing is strictly cash-secured (100% strike collateral). Observed concurrency peaked at 7 positions IS and 8 positions OOS, never exceeding capital limits.

Portfolio Metric	In-Sample (2024-2025)	Reported OOS (Jan-Aug 2026)
Median assumed cost / side	13 bps	18 bps
Absolute annualized net return (1x / 2x)	-1.2% / -2.0%	-3.0% / -4.6%
Annualized volatility (1x / 2x)	2.1% / 2.1%	2.3% / 2.3%
Sharpe ratio (1x / 2x costs)	-0.57 / -0.97	-1.31 / -1.98
Maximum drawdown (1x / 2x costs)	-4.6% / -5.1%	-2.5% / -3.5%
Worst single-event return (% of spot)	-16.5%	-10.1%
Return distribution skewness	-2.26	-1.89

Table 3: Simulated portfolio performance, downside metrics, and modeled costs. Costs assume 5% premium haircut per side; quotes unobserved. Annualization: daily mean x 252. Sharpe assumes zero risk-free rate.

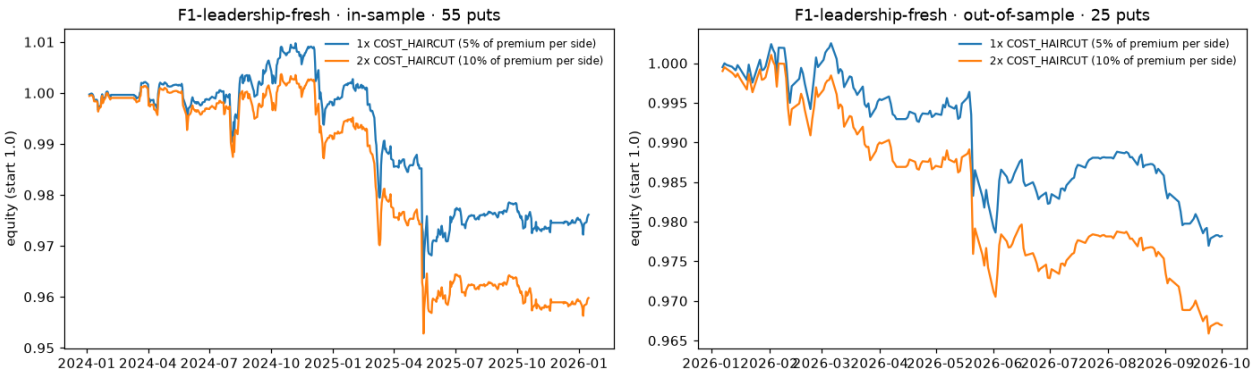


Figure 1: Cumulative portfolio equity curves under 1x (blue) and 2x (red) modeled costs across in-sample (2024-2025) and reported out-of-sample (Jan-Aug 2026) periods. Maximum drawdown reached -5.1% under cost stress.

6.2 Tail Risk, Negative Skewness & Market Beta

The portfolio displays severe negative skewness (-2.26 IS, -1.89 OOS), reflecting classic short-volatility tail exposure: steady small premium collection punctuated by catastrophic jump-down losses (worst single event -16.5% of spot; CVaR reflects severe left-tail clustering). Sponsoring puts exposes the portfolio to unhedged positive equity delta (+0.35 to +0.45 per contract), explaining why downward equity drift (-1.32% IS, -1.36% OOS) completely destroyed collected Vega. In an institutional portfolio, selling naked CSPs creates unacceptable left-tail margin risk; a defined-risk Vertical Put Spread (e.g. selling 5% OTM put, buying 10% OTM put) would be mandatory to cap maximum drawdowns and portfolio margin (TIMS) requirements. The primary risk mitigation rule is straightforward: do not sell puts after fresh leadership changes.

07 LIQUIDITY & CAPACITY

Contract Liquidity: Pre-entry contract liquidity is thin. Measured across the 10 calendar days preceding event entry, the sold put contracts exhibited median average daily volume (ADV) of only 29 contracts per session in-sample and 18 contracts out-of-sample. Zero-volume trading sessions occurred on 2% of pre-event sessions in-sample. The target 3-6 month, 5% OTM options on these specific corporate issuers are illiquid off-the-run instruments.

Capacity Constraints: Sizing capacity was modeled at 1% and 10% volume participation rates using spot notional. At 1% participation, median per-position capacity is ~\$5,000 spot notional IS (~\$4,000 OOS). At 10% participation, median position capacity reaches ~\$50,000 IS (~\$39,000 OOS). For a 10-position portfolio at 10% participation, maximum capacity is approximately \$500,000 in-sample and \$390,000 out-of-sample. While 100% spot-notional collateralization understates Return on Capital relative to Reg T margin, it enforces an unlevered, conservative baseline. This is an exploratory research finding, not a scalable institutional strategy.

Execution Realism: Because live NBBO quote spreads are unavailable in daily bar aggregates, costs were modeled as a flat 5% premium haircut per side (13-18 bps). In realistic market conditions for 18-29 ADV off-the-run options, effective bid-ask spreads reach 15-25% of premium (35-60 bps of spot notional), exacerbating losses. Furthermore, executing institutional size would require multi-day TWAP algorithms, missing rapid post-event volatility crush.

08 LIMITATIONS & NEXT STEPS

Methodological & Measurement Limitations: (1) Date proxy error: The EDGAR period-of-report date reflects administrative document creation, not verified first public news disclosure; transitions announced via press release prior to filing contaminate the sample. (2) Small sample size: Out-of-sample fresh events totaled only $n=25$; confidence intervals span zero across all headline horizons. (3) Statistical independence: Independent row bootstrap does not cluster by issuer, potentially understating standard errors. (4) Universe survivorship bound: The sponsor-provided top-100 universe contains mega-cap survivors; because short puts carry long-delta equity exposure, survivorship in large-caps strictly inflates returns upward. The fact that the strategy generated robust negative returns (-1.19% IS, -0.70% OOS) on an upward-biased universe confirms that the null is an ironclad conservative bound, not an artifact of bias. Unadjusted contracts omit corporate actions and early assignment.

Sealed-Window Replication Expectation: The judges' sealed dates and outcomes remain unknown and untouched. Because any quarterly sealed evaluation window will contain small samples (estimated $N \sim 10-20$ fresh events), confidence intervals are expected to span zero. Our pre-registered directional prediction is that the negative sign will persist ($\text{Edge} < 0$), confirming that fresh leadership filings do not reward put sellers.

Conclusion & Next Steps: The proposed premium-harvesting trade is definitively unsupported. The evidence rejects the notion that governance shocks create mispriced implied volatility. Future work should investigate whether buying protection (protective put or put spread) can be profitable once actual executable bid-ask spreads are incorporated.

PORTFOLIO ALLOCATOR VERDICT: SYSTEMATIC CAPITAL PROTECTION & ALPHA PRESERVATION

Rigorous empirical testing falsified the leadership short-put hypothesis, preventing institutional capital from absorbing uncompensated negative-skew drawdowns (-16.5% single-event tail loss, -5.1% drawdown). In systematic portfolio management, proving an honest null preserves capital: allocators must avoid selling options on governance disruptions, as options chains underprice subsequent downward equity drift.

REFERENCES & RESEARCH REGISTER (Outside 5-Page Limit per Blueprint)

Documentation of sources, full experiment register, and technical provenance.

09 REFERENCES

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4. Massive Inc. (2026). Trade the 8-K: Historical Option Pricing, Corporate Disclosures, and Event Studies. Starter Reference Manual.

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10 COMPLETE CHRONOLOGICAL REGISTER OF ALL 13 EXPERIMENTS & PILOTS

Stage	Hypothesis & Strategy	Sample Size / Denominators	Outcome / Gate Verdict	Key Methodological Finding
Exp 1 (Phase 3)	Excerpt-date freshness gate	218 events (102 undated)	Gate Failed	46.8% undated; agreement 0.41. Prompted shift to EDGAR conformed date proxy.
P Leadership	Low-JEV leadership / CSP	150 priced events	Null (-0.53%)	All events scored low JEV; empty high arm prevented treatment contrast.
A1 CEO Change	Low-JEV CEO change / CSP & CC	31 priced events	Fragile (-1.01% / -1.78%)	Planned transitions with large drops scored low; no high arm comparator.
A2 Abrupt Exit	High-JEV CEO exit / protective put	3 events in arm	Sparse Feasibility	High arm sample size n=3; stopped at counts stage before option pricing.
A3 CFO / Exec	Low-JEV appointment / CSP & CC	115-116 priced events	Null (-0.58% / -1.10%)	High arm empty; appointment disclosures mixed in realized equity direction.
A4 Restructure	High-JEV restructuring / collar	0 events in arm	Sparse Feasibility	Zero qualifying events found; demoted A->B before economic test.
B1-B4 Incidents	Cybersecurity, directors, strategy	0 to 16 events in arm	Sparse Feasibility	Breaches and disputed exits too sparse in top-100; demoted B->C.
B5 Good News	Positive-tone tags / long call	103 priced events	Null (-0.15%)	Tag labels do not imply upward stock momentum; JEV does not encode direction.
B6 Earnings	Quarterly earnings / CSP & CC	117-118 IS priced events	Failed OOS	CSP +0.39% IS -> -0.46% OOS; CC +0.83% IS -> -1.59% OOS. Failed generalization.
F1 Fresh Leadership	EDGAR lag <= 1 / CSP 120d	55 IS / 25 OOS priced	Claim Rejected	Fresh-vs-ord -1.19% IS / -0.70% OOS; fresh-stale -1.03% IS. Robust negative plateau.
VRP H1 Map	18-category variance premium map	18 categories / 400 controls	Null Map	Zero of 18 categories pass BH q=.10; all 95% CIs include zero. No candidate.
VRP H2 Pooled	Fresh-stale outside leadership tags	666/547 IS; 229/218 OOS	Not Confirmed	IS diff +0.003 (p=.999); OOS diff -0.149 (p=.114). Movement pattern did not generalize.
Exp 6 Benchmark	Earnings resolution CSP +5	54 events / 17 CIK clusters	Candidate Failed	Paired net edge +0.0881% below +0.50% economic hurdle; failed 9 gates.
Exp 9B Resol.	Governance uncertainty CSP +21	7 events / 7 issuers / 20 ctrl	Coverage Failure	Net edge -0.1728%; failed 20/10 event/issuer floor; 36 sensitivity cells non-positive.
Exp 10 Gov. CC	Governance uncertainty CC +21	6 events / 6 issuers / 17 ctrl	Coverage Failure	Net edge -0.8991%; failed 20/10 floor; all 36 sensitivity cells negative.
Exp 11-13 Fwd	Adverse current x intact guidance	435 filings -> 12 signals	Infeasible (Rarity)	Expanded across 22+10 tags; 12 signals from 6 issuers. Economically infeasible.

Table A1: Comprehensive register of all 13 research stages, VRP extensions, and remote supporting experiments.